

Problem Set for 12-May and 15-May

The ambient field is $\mathbb{R}$.

Exercise 1 Determine whether the following are inner product spaces (简要回答即可).

1. V is the vector space continuous functions over $\mathbb{R}$, and the pairing is

$$(f, g) \mapsto \int_0^1 f(x)g(x) \, dx.$$

不是内积空间. 存在 $f \neq 0$ 使得 $(f, f) = 0$.

2. $V = \mathbb{R}[x]$, and the pairing is

$$(f, g) \mapsto \int_{-\infty}^{\infty} f(x)g(x)e^{-x^2} \, dx.$$

是内积空间. 显然 (f, g) 是对称双线性型. 对任意 f 总有 $(f, f) < \infty$ (积分收敛). 从而 $(f, g) = \frac{(f+g, f+g) - (f, f) - (g, g)}{2}$ 良定义.

3. $V = \mathbb{R}^{n \times n}$, and the pairing is

$$(A, B) \mapsto \det(A^T \cdot B).$$

不是内积空间 (除去 $n = 1$ 的平凡情况). 存在 $A \neq O$ 使得 $\det(A^T A) = 0$.

4. (**optional**) By axiom of choice, any $\mathbb{R}$ -vector space can be endowed with an inner product structure.

取定义线性空间 V 的一组基 $\mathcal{B}$, 则任意 $x \in V$ 能唯一地写作有限和 $\sum \lambda_i x_i$. 定义正定二次型 $(\sum \lambda_i x_i, \sum \lambda_i x_i) = \sum \lambda_i^2$, 并由此定义内积即可.

Exercise 2 Show that every real matrix is similar to a block diagonal matrix, wherein the candidates of the blocks are

1. the Jordan form of a real eigenvalue;
2. the block of the form

$$\begin{bmatrix} H_{r,\theta} & Q & & & \\ & H_{r,\theta} & Q & & \\ & & \ddots & \ddots & \\ & & & H_{r,\theta} & Q \\ & & & & H_{r,\theta} \end{bmatrix},$$

where

$$1. H_{r,\theta} = \begin{bmatrix} r \cos \theta & -r \sin \theta \\ r \sin \theta & r \cos \theta \end{bmatrix}, \text{ and}$$

$$2. Q = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

对实矩阵在 $\mathbb{C}$ 中的 Jordan 型, 非实数 Jordan 块成对出现. 下直接验证

$$\begin{bmatrix} H_{r,\theta} & Q & & & \\ & H_{r,\theta} & Q & & \\ & & \ddots & \ddots & \\ & & & H_{r,\theta} & Q \\ & & & & H_{r,\theta} \end{bmatrix} \sim \begin{bmatrix} J_n(re^{i\theta}) & O \\ O & J_n(re^{-i\theta}) \end{bmatrix}$$

记 L 为左侧矩阵, R 为右侧矩阵. 依照分块对角矩阵的行列式,

$$\det(\lambda I - L) = (\det(\lambda I - H_{r,\theta}))^n = (x^2 - 2xr \cos \theta + r^2)^n = \det(\lambda I - R).$$

任取 $\mu \in \{re^{\pm i\theta}\}$, 使用待定系数计算得 $\dim \ker(\mu I - L) = 1$. 例如取 $(\mu I - L) \cdot v = 0$,

- 若 v 前两项全为 0, 则 v 的第三与第四项为 0, 归纳得 $v = 0$;
- 若前两项非零, 则比值确定. 取定 v 的前两项, 则 v 的第三与第四项唯一确定.

以上说明 L 在 $\mathbb{C}$ 上的标准型只有两个 Jordan 块, 因此只能是 R 的形式.

Exercise 2.5 Show that every real matrix is a product of two real symmetric matrices.

记 T 是 $\{0, 1\}$ -矩阵, 其中 $T_{i,j} = 1$ 当且仅当 $i + j = n + 1$.

将任意 X 写作上题中的标准型 $P^{-1}LP$, 则 $L = LT \cdot T$ 是对称矩阵的乘积. 因此

$$X = P^{-1}LTP^{-1,T} \cdot P^TLP$$

是两个对称矩阵的乘积.

Exercise 3 Let $(V, (-, -))$ be a finite dimensional real inner product space. For unit vector v_0 (i.e., $(v_0, v_0) = 1$), we define the reflection

$$\varphi : V \rightarrow V, \quad x \mapsto x - 2(x, v_0) \cdot v_0.$$

Show that for isometric transform ψ (i.e., $(\psi(u), \psi(v)) = (u, v)$), 1 is an eigenvalue for ψ or $\psi \circ \varphi$. Overall, $1 \in \sigma(\psi) \cup \sigma(\psi \circ \varphi)$.

本题先猜后证. 关于 x 的构造, 可顺着目标式 $\varphi(x) = \psi(x)$ 画图猜测.

若 1 不是 ψ 的特征值, 则 $(\text{id} - \psi)$ 是同构. 此时, 存在 x 使得 $(\text{id} - \psi)(x) = v_0$. 从几何角度看, $x - \psi(x)$ 与 v_0 同向, 且 ψ 等距, 则应有 $(x + \psi(x), v_0) = 0$. 计算得

$$(x + \psi(x), v_0) = (x + \psi(x), x - \psi(x)) = (x, x) - (\psi(x), \psi(x)) = 0.$$

此时

$$\begin{aligned} \varphi(x) &= x - (2x, v_0)v_0 \\ &= x - (x - \psi(x), v_0)v_0 \\ &= x - (v_0, v_0)v_0 \\ &= x - v_0 \\ &= x - (x - \psi(x)) \\ &= \psi(x). \end{aligned}$$

由 $\varphi^2 = \text{id}$, 得 $1 \in \sigma(\psi \circ \varphi)$.

作为 Euclidean 空间的特例, 即内积形式为 $(u, v) := u^T v$ 的有限维实内积空间, 有以下引理.

Corollary (Minamomo's 1/2 theorem) For orthogonal matrix Q with $1 \notin \sigma(Q)$, the matrix $(I - Q)^{-1}$ has all $\frac{1}{2}$ on its diagonal.

- We present the original proof as follows. For arbitrary index i ,

$$k_i := e_i^T (I - Q)^{-1} e_i \tag{1}$$

is exactly the (i, i) -th entry of $(I - Q)^{-1}$. We see that

$$k_i = e_i^T (I - Q^T)^{-1} (Q^{-1} Q) e_i = e_i^T (I - Q)^{-1} (-Q) e_i. \tag{2}$$

Hence, (1) + (2) yields

$$2k_i = e_i^T (I - Q)^{-1} (I - Q) e_i = 1.$$

The theorem also holds for unitary matrices.

- Historical remark: Minamomo is one of our classmates who discovered this amazing theorem last night, where Minamomo is his/her wechat name written in Hiragana.

Exercise 4 Set $V := \mathbb{R}[x]$ and $V_0 := \{f \in \mathbb{R}[x] \mid f(0) = f(1)\}$.

1. Prove that $V \times V \rightarrow \mathbb{R}$, $(f, g) \mapsto \int_0^1 f(x)g(x) dx$ is an inner product.
2. Set $\mathcal{D} : V_0 \rightarrow V$, $f(x) \mapsto f'(x)$. Find $\dim \ker(\mathcal{D})$ and $\dim \operatorname{coker}(\mathcal{D})$.
3. Define the inner product restricted on the subspace

$$(\cdot, \cdot)_0 : V_0 \times V_0 \rightarrow \mathbb{R}, \quad (f, g) \mapsto \int_0^1 f(x)g(x) dx.$$

Is there any linear map $\mathcal{D}^* : V \rightarrow V_0$ such that for any $h \in V_0$ and $g \in V$,

$$(\mathcal{D}^* g, h)_0 = (g, \mathcal{D}h)?$$

内积的对称双线性性是显然的, 关键是说明 $(f, f) = 0 \implies (f = 0)$. 若多项式在 $(0, 1)$ 上恒零, 则自然是零多项式.

记 $\mathcal{D} = D \circ i$, 其中 $i : V_0 \rightarrow V$ 是包含映射, D 是 V 上的导数.

1. 由 i 是单射, 故 $\dim \ker Di \leq \dim \ker D = 1$, 得 $\dim \ker Di \in \{0, 1\}$. 由常多项式属于 V_0 , 取 1.
2. 由 D 是满射, 故 $\dim \operatorname{coker} Di \leq \dim \operatorname{coker} i = 1$, 得 $\dim \operatorname{coker} Di \in \{0, 1\}$. 由常多项式不属于 $\operatorname{im}(Di)$, 取 1.

不存在. 下使用反证法. 若 $\mathcal{D}^*$ 存在, 对任意 h ($h(0) = h(1) = 0$) 与 $g \in V$, 总有

$$(\mathcal{D}^* g, h)_0 = (g, \mathcal{D}h) = (-g', h).$$

取 $V_1 = \{x(x-1)f \mid f \in V\}$ 为以上 h 所在的线性空间, 因此 $(\mathcal{D}^* g + g') \in V_1^\perp$. 对任意 $\varphi \in V_1^\perp$, 有

$$0 = (\varphi, x(x-1)\varphi) = \int_0^1 x(x-1)\varphi^2 dx.$$

因此 $\varphi = 0$. 这说明 $V_1^\perp = 0$, 从而 $(\mathcal{D}^* g + g') = 0$. 这与 $\operatorname{im}(\mathcal{D}^*) \subseteq V_0$ 矛盾.

Exercise 5 Let $(V, (\cdot, \cdot))$ be an inner product space, where V is not necessary finite dimensional.

- Say φ^* is the adjoint of $\varphi \in \mathbf{Hom}_{\mathbb{R}}(V, V)$, provided $(\varphi^*(x), y) = (x, \varphi(y))$ for arbitrary $x, y \in V$.
- Let U be a subspace of V . Set

$$U^\perp := \{v \in V \mid (u, v) = 0, \forall u \in U\} = \bigcap_{u \in U} \ker((u, -))$$

Now we assume that φ is invertible, and φ^* exists.

1. Show that φ^* is injective, and $(\text{im}(\varphi^*))^\perp = 0$.

即证 $[\varphi^*(y) = 0] \implies [y = 0]$. 由 φ 可逆, 取原像 $x = \varphi^{-1}(y)$, 得

$$0 = (\varphi^*(y), x) = (y, \varphi(x)) = (y, y).$$

由内积空间定义, $y = 0$.

再证 $z \in \text{im}(\varphi^*)^\perp$ 必为 0. 此时 $(\varphi^*(-), z) = (-, \varphi(z))$ 是零映射, 因此 $\varphi(z) = 0$, 故 $z = 0$.

2. Show that φ^* is surjective $\implies (\varphi^{-1})^*$ exists and $(\varphi^{-1})^* = (\varphi^*)^{-1}$.

结合上一问, 题设是 φ^* 为同构. 此时

$$(x, \varphi^{-1}(y)) = ((\varphi^*)(\varphi^*)^{-1}(x), \varphi^{-1}(y)) = ((\varphi^*)^{-1}(x), \varphi(\varphi^{-1}(y))) = ((\varphi^*)^{-1}(x), y).$$

这一构造表明 $(\varphi^*)^{-1} = (\varphi^{-1})^*$.

3. Show that $(\varphi^{-1})^*$ exists $\implies \varphi^*$ is invertible and $(\varphi^{-1})^* = (\varphi^*)^{-1}$.

若 $(\varphi^{-1})^*$ 存在, 则

$$((\varphi^{-1})^*(\varphi^*)(x), y) = (\varphi^*(x), \varphi^{-1}(y)) = (x, \varphi(\varphi^{-1}(y))) = (x, y).$$

这说明 $(\varphi^{-1})^*\varphi^*$ 是恒等映射. 类似的计算 (调换位置) 表明 $\varphi^*(\varphi^{-1})^*$ 是恒等映射, 从而 φ^* 与 $(\varphi^{-1})^*$ 是互逆的.

4. Let $V = \mathbb{R}[x]$ and $(f, g) := \sum_{n \geq 0} f_n g_n$, where $h = \sum h_i \cdot x^i$. Show that

1. $(V, (,))$ is an inner product space;
2. $L : f \mapsto \frac{f - f(0)}{x}$ is the linear map of moving left. Show that $(\text{id} + L)$ is invertible;
3. Show that $\varphi := (\text{id} + L)^{-1}$ has no adjoint φ^* .

这都是直接的验证. 对于最后一点, 若 $((\text{id} + L)^{-1})^*$ 存在, 则 $(\text{id} + L)^*$ 是可逆的映射. 直接的计算表明 $(\text{id} + L)^* = (\text{id} + R)$ 非满, 因为 $x \in \text{im}(\text{id} + R)$ 必然满足 $\sum x_i = 0$.

Exercise 6 (Optional, very interesting). Find the dimension and inertia of the following symmetric bilinear form, and verify the invariant group action.

1. $V = \{X \in \mathbb{R}^{2 \times 2} \mid \text{tr}(X) = 0\}$, and the pairing is $(A, B) \mapsto \text{tr}(A \cdot B)$. Moreover, for $g \in \text{GL}_2(\mathbb{R})$, the pairing is invariant under $X \mapsto g \cdot X \cdot g^{-1}$.

Step 0 $\dim V = 3$.

Step 1 The pairing is linear (trivial) and symmetric, since $\text{tr}(AB) = \text{tr}(BA)$.

Step 2 Let $(A_1, A_2, A_3) := (E_{1,2}, E_{2,1}, E_{1,1} - E_{2,2})$ be a basis of V , the associated Gram matrix is the 3×3 symmetric matrix $(\text{tr}(A_i A_j))_{3 \times 3}$ is $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$. The inertia is $(++-)$.

Step 3 Since $\text{tr}((gXg^{-1}) \cdot (gYg^{-1})) = \text{tr}(XY)$, pairing is invariant under such group action.

2. $V = \mathbb{R}^{2 \times 2}$, and the pairing is $(A, B) \mapsto \text{tr}(\text{adj}(A) \cdot B)$. Moreover, for $(g, h) \in \text{SL}_2(\mathbb{R}) \times \text{SL}_2(\mathbb{R})$, the pairing is invariant under $X \mapsto g \cdot X \cdot h^{-1}$.

$\dim V = 4$. 基 $(E_{1,1} \pm E_{2,2}, E_{1,2} \pm E_{2,1})$ 对应惯性指数 $(\pm, \pm)$. 对 $g \in \text{SL}_2$, $\text{adj}(g) = g^{-1}$.

$$\text{tr}(\text{adj}(gXh^{-1})gYh^{-1}) = \text{tr}(h\text{adj}(X)g^{-1}gYh^{-1}) = \text{tr}(\text{adj}(X)Y).$$

3. $V = \left\{ \begin{pmatrix} a & ib \\ ic & \bar{a} \end{pmatrix} \mid a \in \mathbb{C}; b, c \in \mathbb{R} \right\}$, the pairing is $(X, Y) \mapsto \text{Re}(\text{tr}(X \cdot \bar{Y}))$. Moreover, for $g \in \text{GL}_2(\mathbb{C})$, the pairing is invariant under $X \mapsto g \cdot X \cdot (\bar{g})^{-1}$.

$\dim V = 4$. 基 $(I, i(E_{1,1} - E_{2,2}), i(E_{1,2} \pm E_{2,1}))$ 对应惯性指数 $(+, +, \pm)$. 对 $g \in \text{GL}_2(\mathbb{C})$,

$$\text{Re}(\text{tr}(gX\bar{g}^{-1} \cdot \overline{gY\bar{g}^{-1}})) = \text{Re}(\text{tr}(X \cdot \bar{Y})).$$

4. (**Optional**, if you know tensor product). Let $U \wedge U$ be the quotient space of $U \otimes U$ by the relation $(a \otimes b + b \otimes a = 0)$. In particular, $\dim(U \wedge U) = \binom{\dim U}{2}$. Let $V = \mathbb{R}^4 \wedge \mathbb{R}^4$ be the 6-dimensional vector space with basis $\{e_i \wedge e_j\}_{1 \leq i < j \leq 4}$, and the pairing is $(\lambda, \mu) \mapsto \lambda \wedge \mu$. Here $\bigwedge_{i=1}^4 \mathbb{R}^4$ is one-dimensional with basis $\{e_1 \wedge e_2 \wedge e_3 \wedge e_4\}$. Moreover, for $g \in \mathrm{SL}_4(\mathbb{R})$, the pairing is invariant under $\mu \mapsto g \cdot \mu$. Here $g \cdot (\sum c_{i,j} e_i \wedge e_j) := \sum c_{i,j} (g \cdot e_i) \wedge (g \cdot e_j)$.

$\dim V = 6$. 基

$$(e_1 \wedge e_2 \pm e_3 \wedge e_4, e_1 \wedge e_3 \pm e_2 \wedge e_4, e_1 \wedge e_4 \pm e_2 \wedge e_3)$$

对应惯性指数 $(\pm, \mp, \pm)$. 对 $g \in \mathrm{SL}_4(\mathbb{R})$,

$$\det(g \cdot \sum u_i \wedge v_i, g \cdot \sum x_j \wedge y_j) = \sum_{i,j} \det(gu_i \wedge gv_i \wedge gx_j \wedge gy_j).$$

考虑 $\det ge_1 \wedge ge_2 \wedge ge_3 \wedge ge_4 = (\det g) \cdot (\det e_1 \wedge e_2 \wedge e_3 \wedge e_4)$, 上式 g 可去.

The above calculation provides the elementary construction of low dimensional spin groups, that is, the double covering of special orthogonal groups. One learns from these construction that

1. $\mathrm{SL}_2(\mathbb{R})/\{\pm I\} \simeq \mathrm{SO}(2, 1)$, and $\mathrm{SL}_2(\mathbb{C})/\{\pm I\} \simeq \mathrm{SO}_3(\mathbb{C})$;
2. $(\mathrm{SL}_2(\mathbb{R}))^2/\{\pm(I, I)\} \simeq \mathrm{SO}(2, 2)$, and $(\mathrm{SL}_2(\mathbb{C}))^2/\{\pm(I, I)\} \simeq \mathrm{SO}_4(\mathbb{C})$;
3. $\mathrm{SL}_2(\mathbb{C})/\{\pm I\} \simeq \mathrm{SO}(3, 1)$;
4. $\mathrm{SL}_4(\mathbb{C})/\{\pm I\} \simeq \mathrm{SO}_6(\mathbb{C})$, and $\mathrm{SL}_4(\mathbb{R}) \simeq \mathrm{SO}(3, 3)$.

As corollaries, $\mathrm{SO}_4(\mathbb{C})$ is a double cover of $\mathrm{SO}_3(\mathbb{C})$, and the group isomorphism $\mathrm{SO}(3, 1) \simeq \mathrm{SO}_3(\mathbb{C})$ is a famous coincidence in quantum physics.